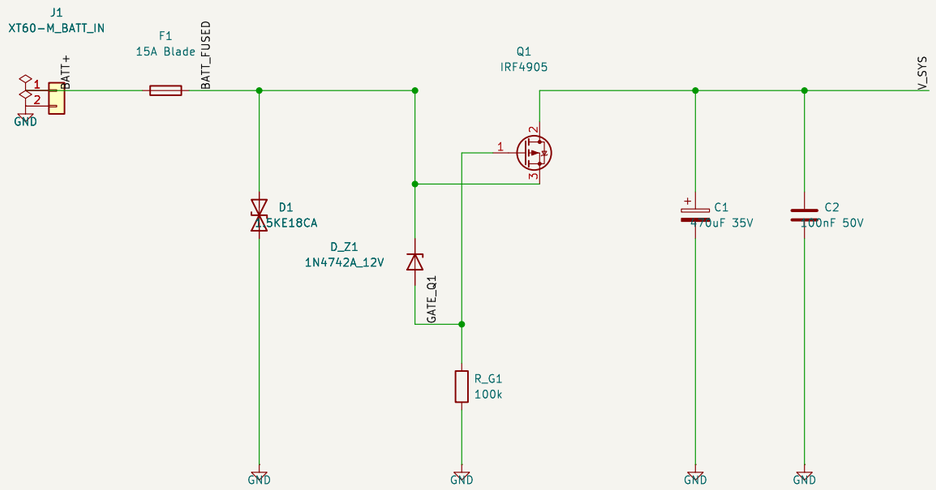
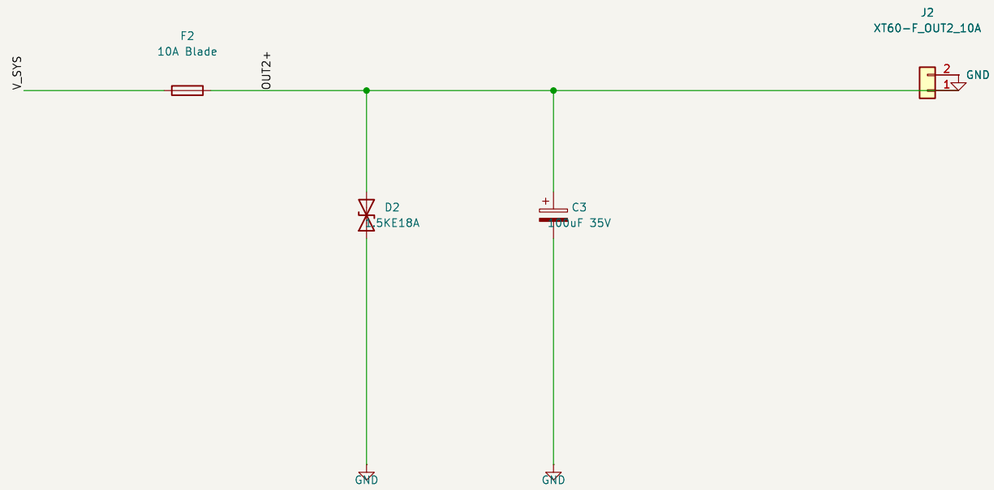
STAGE 1: PRIMARY INPUT HARDENING & REVERSE POLARITY (10.0V – 16.8V)

15A Blade Fuse | 1.5KE18CA TVS | IRF4905 P-FET | 1N4742A Zener Clamp



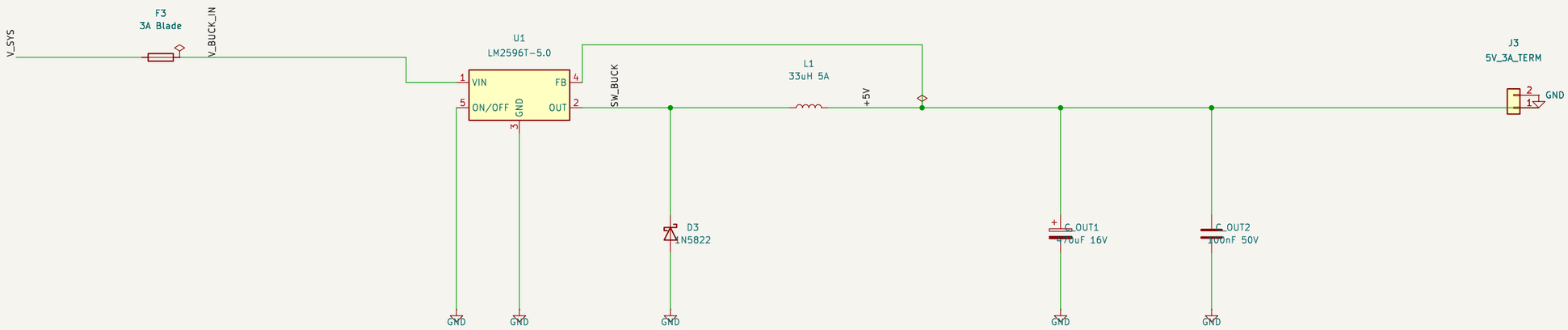
STAGE 2: PROTECTED PASS-THROUGH OUTPUT 2 (10A CONT, 10.0V – 16.8V)

10A Automotive Blade Fuse | 1.5KE18A TVS | 100uF Low-ESR Filter



STAGE 3: REGULATED 5V @ 3A STEP-DOWN BUCK REGULATOR (150kHz)

3A Blade Fuse | LM2596T-5.0 TO-220-5 | 1N5822 Schottky | 33uH 5A Toroid



Sheet: /		
File: battery_protection_board_tht.kicad_sch		
Title: 4S Battery Protection & Power Distribution Board		
Size: A3	Date: 2026-10-22	**Rev: v1.1**
KiCad E.D.A. 9.0.9.1		Id: 1/1